


$$Z \sim N(0, 1)$$

$$X = \mu + \sigma Z \quad X \sim N(\mu, \sigma^2)$$

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

$$= \frac{\sum_i (x_i - \bar{x}) y_i}{s_{xx}}$$

$$\delta_i = \frac{(x_i - \bar{x})}{s_{xx}} = \sum_i \delta_i y_i \quad /$$

$$\sum_i [(x_i - \bar{x}) y_i - (x_i - \bar{x}) \bar{y}]$$

$$= \sum_i (x_i - \bar{x}) y_i$$

$$\sum_i (x_i - \bar{x}) \bar{y}$$

$$A5 \quad Y_i | X \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$$

$$A1: E[Y_i | X_i] = \beta_0 + \beta_1 X_i$$

Practice Midterm

Problem 3:

$$(d) \quad \widehat{wage} = \hat{\beta}_0 + \hat{\beta}_1 (120) \rightarrow \text{using estimates number}$$

$$(e) \quad E[\widehat{wage} | Q's] = E[\hat{\beta}_0 + \hat{\beta}_1 (120) | Q's] \\ = E[\hat{\beta}_0 | Q's] + 120 E[\hat{\beta}_1 | Q's]$$

To estimate those two moments

$$\begin{aligned} \widehat{wage} &= \hat{\beta}_0 + 120 \hat{\beta}_1 \\ \text{Var}(\widehat{wage} | Q's) &= \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 (120) | Q's) \\ &= \text{Var}(\hat{\beta}_0 | Q's) + 120^2 \text{Var}(\hat{\beta}_1 | Q's) \\ &\quad + 2(120) \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 | Q's) \end{aligned}$$

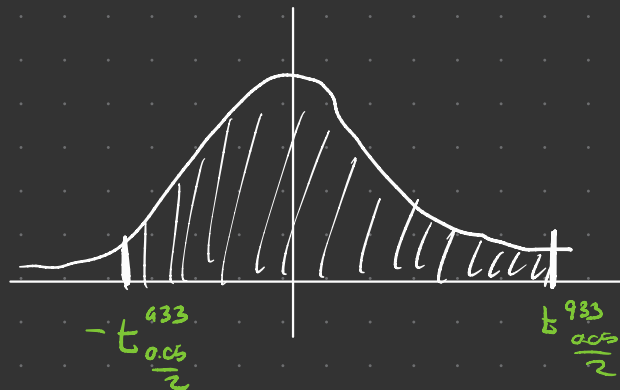
$$\hat{wage} = \hat{\beta}_0 + \hat{\beta}_1 (120) \quad E(\hat{wage} | Q's) = \beta_0 + \beta_1 (120) = \mu_{wage}$$

$$Z = \frac{\hat{wage} - \mu_{wage}}{\sqrt{\text{Var}(\hat{wage})}} \sim N(0, 1)$$

$$T = \frac{\hat{wage} - \mu_{wage}}{\sqrt{\hat{\text{Var}}(\hat{wage})}} \sim t_{\text{dist}}(935 - 2)$$

$$= \frac{\hat{wage} - \mu_{wage}}{SE(\hat{wage})}$$

$$Pr\left(\left|\frac{\hat{wage} - \mu_{wage}}{SE(\hat{wage})}\right| \leq t_{\frac{0.05}{2}}^{433}\right) = .95$$



$$Pr\left(\hat{wage} - t_{\frac{0.05}{2}}^{433} \cdot SE(\hat{wage}) \leq \mu_{wage} \leq \hat{wage} + t_{\frac{0.05}{2}}^{433} \cdot SE(\hat{wage})\right) = 0.95$$

$$CI = \hat{wage} \pm t_{\frac{0.05}{2}}^{433} \cdot SE(\hat{wage})$$

$$\frac{\bar{X} - \mu}{\frac{\sqrt{\text{Var}(X)}}{\sqrt{n}}} \xrightarrow{d} N(0,1) \quad \text{CLT}$$

$$\bar{X} \xrightarrow{P} E[X] \quad \text{WLLN}$$

$f(x; \theta)$ we are going to estimate θ with $\hat{\theta}_n$

$$\hat{\theta}_n \xrightarrow{P} \theta ;$$

$$1) \quad Z_n = \frac{\hat{\theta}_n - \theta}{\sqrt{\frac{\text{Var}(\hat{\theta}_n)}{n}}} \xrightarrow[d]{\text{CLT}} N(0,1) \quad \text{where} \quad \text{Var}(X) = \sigma^2$$

$$\text{Var}(\hat{\theta}_n) = \frac{\sigma^2}{n}$$

$$2) \quad \text{Propose} \quad \hat{\text{Var}}(\hat{\theta}_n) \xrightarrow{P} \text{Var}(\hat{\theta}_n)$$

$$3) \quad W_n = \frac{\hat{\text{Var}}(\hat{\theta}_n)}{\text{Var}(\hat{\theta}_n)} \xrightarrow{P} 1$$

$$4) \quad T_n = \frac{\hat{\theta}_n - \theta}{\sqrt{\frac{\hat{\text{Var}}(\hat{\theta}_n)}{n}}} = \frac{Z_n}{W_n} \xrightarrow[d]{\text{Slutsky's}} N(0,1)$$

A1 - A5 True

$$Z = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\text{Var}(\hat{\beta}_1)}} = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{s_{xx}}}} \sim N(0, 1)$$

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum \hat{\varepsilon}_i^2 \quad \frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-2)$$

$\hat{\sigma}^2$ estimates σ^2

$$T = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\hat{\sigma}^2}{s_{xx}}}}$$

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{\sigma^2}{s_{xx}}}} \sim t(n-2)$$
$$\sqrt{\frac{(n-2)\hat{\sigma}^2}{\sigma^2}} \sim t(n-2)$$

$$t_{\text{dist}} = \frac{z}{\sqrt{\frac{s^2}{\text{degrees}}}}$$